

Quadratic Witt Classes over $\mathbb{F}_2(t)$: Residue Coordinates, Impartial Realization, and a Singular Obstruction

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Abstract

The quadratic Witt group of the imperfect field $K = \mathbb{F}_2(t)$ carries wild local information absent from the finite-field Arf classification. Using the characteristic-two Milnor–Scharlau sequence of Aravire and Jacob, we prove that a class is determined canonically by its unramified coordinate at infinity together with all second residues. At a place v , the second residue consists of a finite odd-pole expansion over the residue field and one ramified Arf bit. We prove directly that every nonzero finite-field Scharlau transfer preserves the Arf generator. Consequently the single global relation is the parity of the ramified bits.

Finite-field trace characters turn every wild coefficient into a basis-free finite family of binary observations. Sending zero to the zero game and one to $*$ gives a faithful, orthogonal-sum-additive family of impartial normal-play values for every nonsingular Witt class. This is a canonical classification-to-game compiler, not a refinement-independent local-access arena. We identify ramified Artin–Schreier coordinate change as the exact remaining naturality problem. Finally, we prove that no additive impartial invariant can extend faithfully to arbitrary singular quadratic spaces: the quasilinear relation $\langle c \rangle \perp \langle c \rangle \cong \langle c \rangle \perp \langle 0 \rangle$ forces cancellation of the radical coefficient.

1 Introduction

Over a finite field of characteristic two, the quadratic Witt group is classified by one Arf bit [3, 5]. This finite situation admits an exact normal-play interpretation: after restriction of scalars to $\mathbb{F}_2$, a binary quadratic refinement can be compiled into an impartial arena whose losing loadings are its zero set [1]. Neither statement extends naively to $K = \mathbb{F}_2(t)$. The field is imperfect, and odd negative Laurent powers survive Artin–Schreier reduction at every place. The resulting wild terms are genuine Witt coordinates, not implementation noise.

Aravire and Jacob construct characteristic-two second residues and transfers and prove the exact sequence [2, Theorem 6.2]

$$0 \longrightarrow W_q(\mathbb{F}_2) \longrightarrow W_q(K) \xrightarrow{\oplus_v \partial_v} \bigoplus_v W_1(K_v) \xrightarrow{\oplus_v s_v^*} W_q(\mathbb{F}_2) \longrightarrow 0.$$

The notation is important: $W_1(K_v)$ is a quotient of the Witt group of the completion, not the Witt group of its residue field. Its wild summand is infinite-dimensional, although every individual residue has finite support.

This paper extracts the exact normal-play consequence of that sequence and separates it from two stronger requests: constructing the coordinates without first running a Witt classifier, and retaining arbitrary quasilinear radical data under an additive game law.

Theorem A. Fix $K = \mathbb{F}_2(t)$ and the canonical uniformizers $p(t)$ at finite places and t^{-1} at infinity.

1. There is a canonical additive isomorphism

$$W_q(K) \cong \mathbb{F}_2 \oplus \left\{ ((\psi_v, \epsilon_v))_v : \text{finite support and } \sum_v \epsilon_v = 0 \right\},$$

where ψ_v is the wild odd-pole residue and ϵ_v the ramified Arf bit.

2. These coordinates have a finite, basis-free, faithful realization by the impartial values 0 and *, additive under orthogonal sum.
3. No isometry-invariant orthogonal-sum homomorphism into a cancellative game-value monoid can distinguish every quasilinear radical restriction.

The first item uses the cited Milnor–Scharlau exact sequence. The finite-field transfer calculation, coordinate splitting, game realization, and singular obstruction are proved here.

2 Local characteristic-two coordinates

Let v range over the monic irreducible polynomials $p \in \mathbb{F}_2[t]$ and the infinite place. Define

$$\pi_p = p(t), \quad \pi_\infty = t^{-1}.$$

Write K_v for the completion and κ_v for its residue field. Thus

$$\kappa_p = \mathbb{F}_2[t]/(p), \quad \kappa_\infty = \mathbb{F}_2.$$

Every κ_v is finite and therefore perfect. The local theorem of Aravire and Jacob specializes to [2, Theorem 1.3 and Corollary 1.7]

$$W_q(K_v) \cong W_q(\kappa_v) \oplus R_v \oplus \langle \pi_v \rangle W_q(\kappa_v),$$

where

$$R_v = \bigoplus_{\substack{n \geq 1 \\ n \text{ odd}}} \kappa_v \pi_v^{-n}.$$

The first summand is unramified. Quotienting it out gives

$$W_1(K_v) := \text{coker}(W_q(\kappa_v) \longrightarrow W_q(K_v)) \cong R_v \oplus \mathbb{F}_2.$$

For a global class α , write its localization as

$$\text{loc}_v(\alpha) = (\phi_{0,v}, \psi_v, \phi_{1,v}), \quad \psi_v = \sum_{n \text{ odd}} c_{v,n} \pi_v^{-n}.$$

Then

$$\partial_v \alpha = (\psi_v, \phi_{1,v}).$$

Only finitely many places occur for one global class, and each ψ_v is a finite sum. The uniformizer is part of this description: changing it changes the displayed odd-pole coordinates even when the underlying class is fixed.

3 Finite-field Scharlau transfer

We now identify the final map in the global sequence. The proof uses only the nondegenerate trace pairing of a finite field extension [6].

Lemma 3.1 (Transfer preserves the Arf generator). *Let $E/\mathbb{F}_2$ be finite and let $s : E \rightarrow \mathbb{F}_2$ be a nonzero linear functional. Under the Arf identifications*

$$W_q(E) \cong \mathbb{F}_2 \cong W_q(\mathbb{F}_2),$$

the Scharlau transfer $s_ : W_q(E) \rightarrow W_q(\mathbb{F}_2)$ is the identity.*

Proof. The trace pairing is nondegenerate, so $s(z) = \text{Tr}(cz)$ for a unique $c \neq 0$. Since E is perfect, choose $d \in E$ with $d^2 = c$. Let $a \in E$ satisfy $\text{Tr}(a) = 1$; the plane $[1, a]$ represents the nonzero class in $W_q(E)$. Its transfer along s is

$$s(x^2 + xy + ay^2) = \text{Tr}(cx^2 + cxy + cay^2).$$

After the simultaneous change of variables $X = dx, Y = dy$, this is

$$T_a(X, Y) = \text{Tr}(X^2 + XY + aY^2).$$

Choose a basis (e_i) of $E/\mathbb{F}_2$ and its trace-dual basis (f_i) . The vectors $(e_i, 0), (0, f_i)$ form a symplectic basis for the polar form of T_a . Therefore

$$\begin{aligned} \text{Arf}(T_a) &= \sum_i \text{Tr}(e_i^2) \text{Tr}(af_i^2) \\ &= \text{Tr}\left(a \sum_i \text{Tr}(e_i)f_i^2\right). \end{aligned}$$

Trace duality applied to $1 \in E$ gives

$$1 = \sum_i \text{Tr}(e_i)f_i.$$

Squaring in characteristic two gives $1 = \sum_i \text{Tr}(e_i)f_i^2$. Hence

$$\text{Arf}(T_a) = \text{Tr}(a) = 1.$$

The nonzero class is therefore carried to the nonzero class. □

In the Aravire–Jacob construction, s_v^* kills R_v and applies ordinary Scharlau transfer to the ramified residue form. Lemma 3.1 therefore gives

$$s_v^*(\psi_v, \phi_{1,v}) = \phi_{1,v}.$$

Thus wild coefficients are locally essential but globally unconstrained; the only reciprocity relation is parity of the ramified bits.

4 Canonical global coordinates

The exact sequence determines classes modulo constants. A distinguished unramified local coordinate supplies the missing constant without choosing a vector-space basis.

Theorem 4.1 (Infinity plus second residues). *Define*

$$\Theta(\alpha) = (\phi_{0,\infty}(\alpha), ((\psi_v(\alpha), \phi_{1,v}(\alpha)))_v).$$

Then Θ is an additive isomorphism

$$W_q(\mathbb{F}_2(t)) \xrightarrow{\sim} \mathbb{F}_2 \oplus \left\{ ((\psi_v, \epsilon_v))_v : \text{finite support and } \sum_v \epsilon_v = 0 \right\}.$$

Proof. The transfer formula above identifies the kernel of the final arrow in the Milnor–Scharlau sequence with the displayed parity-zero residue tuples. The unramified coordinate $\phi_{0,\infty}$ is additive. On a constant-field class $a \in W_q(\mathbb{F}_2)$, localization at infinity has coordinates $(a, 0, 0)$, so $\phi_{0,\infty}$ restricts to the identity on constants.

Suppose $\Theta(\alpha) = 0$. All second residues vanish, so exactness gives α in the image of $W_q(\mathbb{F}_2)$. Its infinite unramified coordinate is that constant class, and hence the first coordinate makes it zero. This proves injectivity.

Conversely, let r be a finite residue tuple of parity zero. Exactness gives $\beta \in W_q(K)$ with residue r . Given a prescribed constant coordinate $a \in \mathbb{F}_2$, set

$$\alpha = \beta + (a - \phi_{0,\infty}(\beta)),$$

where the parenthesized term is included as a constant class. It changes no second residue and changes the first coordinate to a . Thus Θ is surjective. Injectivity makes the lift unique. $\square$

Remark 4.2. Exactness alone gives a noncanonical splitting because these groups are $\mathbb{F}_2$ -vector spaces. The theorem is stronger: the first local coordinate at the fixed infinite place is an explicit retraction, so no basis extension or choice of residue lifts appears in Θ .

5 A finite impartial realization

The wild coefficient $c_{v,n} \in \kappa_v$ is not itself a bit. Expanding it in a power basis would be finite but would introduce a coordinate choice. The trace pairing gives a redundant basis-free presentation instead.

Definition 5.1 (Trace-bit presentation). For $z \in \kappa_v$, define

$$b_{v,n,z}(\alpha) = \text{Tr}_{\kappa_v/\mathbb{F}_2}(zc_{v,n}).$$

Also put

$$b_c(\alpha) = \phi_{0,\infty}(\alpha), \quad b_{v,r}(\alpha) = \phi_{1,v}(\alpha).$$

Each bit is additive in α . Nondegeneracy of the trace pairing says that $c_{v,n} = 0$ exactly when every $b_{v,n,z}$ vanishes. The complete bit family is faithful by Theorem 4.1. It also has finite support: one class has finitely many places and pole orders, and each κ_v is finite.

Let 0 denote the optionless impartial game and let $*$ = $\{0 \mid 0\}$. These are the numbers 0 and 1, so $* + * = 0$ under disjunctive sum [4].

Theorem 5.2 (Sparse impartial compiler). *For every nonsingular quadratic Witt class α over $\mathbb{F}_2(t)$, replace each zero trace, constant, or ramified bit by the zero game and each one bit by $*$. Omit zero entries from the resulting labelled family. Then:*

1. *the family is finite and a coordinate is a P-position exactly when its bit is zero;*
2. *orthogonal sum becomes coordinatewise disjunctive sum;*
3. *hyperbolic classes map to the zero family; and*
4. *two families have the same outcome vector exactly when their source forms are Witt equivalent.*

Proof. Finiteness and faithfulness were proved above. The map $\mathbb{F}_2 \rightarrow \{0, *\}$ is an additive injection because $* + * = 0$. Applying it to every labelled bit proves the orthogonal-sum identity. Theorem 4.1 then identifies equality of all bits with equality in $W_q(K)$; the zero class consists precisely of hyperbolic forms. $\square$

This construction is not the Boolean serialization of one displayed rational function. Every bit is a homomorphism defined on the Witt group after passage through the exact residue sequence, so it is invariant under isometry and hyperbolic stabilization. It is nevertheless a classifier-to-star compiler: the residue has been computed before the one-move game is emitted.

6 Naturality and the local-access frontier

The trace-bit presentation is automatically natural under unramified residue extension. Let ℓ/κ be finite and let $\iota : \kappa \rightarrow \ell$ be the inclusion. Trace transitivity gives

$$\mathrm{Tr}_{\ell/\mathbb{F}_2}(z' \iota(c)) = \mathrm{Tr}_{\kappa/\mathbb{F}_2}(\mathrm{Tr}_{\ell/\kappa}(z')c).$$

The adjoint formula with $c' \in \ell$ similarly makes residue-field transfer commute with the bit presentation. Thus the redundant trace labels are an advantage: naturality is a relabelling through relative trace, not a matrix identity between chosen bases.

Ramified scalar extension is the exact unresolved case. If

$$\pi_v = u\varpi_w^e,$$

then an odd pole becomes

$$c\pi_v^{-n} = cu^{-n}\varpi_w^{-en}.$$

When en is even, Artin–Schreier normalization removes the leading pole and can feed lower odd poles and the unramified coordinate. The resulting transformation is triangular in pole order but is not coordinatewise extension of residue scalars. A full base-change theorem must prove this normalization law and its compatibility with the local transfer diagrams.

There is a second constructive frontier. The compiler of Theorem 5.2 precomputes each invariant. A Gold-style local-access theorem would instead require a refinement-independent board and bounded diagonal queries during play. The existing weighted-source Witt–FIFO arena provides that stronger access discipline for one finite binary refinement [1]. Extending its internal accounting to the complete local odd-pole normal form, while retaining the trace-adjoint identity above and the ramified transformation, is the remaining game construction.

7 The Gold–Arf boundary

The distinction between pointwise quadratic realization and Witt realization is structural, not merely an implementation choice.

Proposition 7.1 (Pointwise loadings do not descend). *A Witt-invariant additive family cannot literally reproduce the complete loaded-root family of a pointwise quadratic realization. It can recover only its Witt/Arf shadow.*

Proof. The hyperbolic plane

$$H(x, y) = xy$$

is zero in $W_q(\mathbb{F}_2)$. Every Witt-invariant realization must therefore map H to the zero family. But $H(1, 1) = 1$, so any exact pointwise realization has an N -position at the loading $(1, 1)$. The pointwise outcome function is not invariant under hyperbolic stabilization. $\square$

For a nonsingular finite binary form, the Arf invariant controls the sign of the zero-set bias. This class-level statistic is exactly what the constant coordinate of Theorem 4.1 retains. Thus the compatible meaning of “recover Gold–Arf” is agreement of the Witt coordinate and outcome bias, not literal equality of all loaded boards.

8 A singular additive obstruction

The nonsingular Witt group is cancellative. Arbitrary singular quadratic spaces form a different object: quasilinear directions already make their orthogonal-sum monoid noncancellative.

Theorem 8.1 (Quasilinear cancellation obstruction). *Let F be a field of characteristic two. No isometry-invariant, orthogonal-sum-additive map from all finite-dimensional quadratic spaces over F to a cancellative commutative monoid can distinguish the restrictions of all forms to their polar radicals. In particular, no such family of impartial game values is faithful on arbitrary singular spaces.*

Proof. Fix $c \neq 0$ and write $L_c = \langle c \rangle$ and $Z = \langle 0 \rangle$. The invertible change of variables $(u, v) = (x + y, y)$ gives

$$L_c \perp L_c \cong L_c \perp Z, \quad cx^2 + cy^2 = c(x + y)^2.$$

Let f be an additive isometry invariant with values in a cancellative commutative monoid. Then

$$f(L_c) + f(L_c) = f(L_c) + f(Z).$$

Cancellation gives $f(L_c) = f(Z)$. But the restrictions to the polar radicals are respectively the nonzero quasilinear form cx^2 and the zero form. Therefore f cannot retain the requested radical datum.

Finite impartial normal-play values are nimbers, and disjunctive sum is their xor addition, so their value monoid is cancellative. The conclusion applies coordinatewise to any additive impartial family. $\square$

The obstruction leaves two coherent alternatives. One may retain orthogonal-sum additivity and classify only the cancellative quotient of the quasilinear monoid, or retain the full radical restriction in a separate nonadditive report. Requiring both is inconsistent.

9 Formal verification and implementation boundary

The Ogdoad Lean module `WittRealizationExact` formalizes the parts of the proof that do not require a quadratic Witt-group library [1]. Given linear maps fitting the exact-sequence hypotheses, it proves:

- the residue image is exactly the transfer kernel;
- equal residues differ by a unique constant;
- an explicit constant detector gives the linear equivalence $W \simeq C \times \ker(\text{transfer})$;
- $z \mapsto (\lambda \mapsto \text{Tr}(\lambda z))$ is a basis-free injective finite-field presentation; and
- an absorption relation $a + a = a + z$ defeats every additive injection into a cancellative target.

Lean does not formalize the Aravire–Jacob sequence, the local quadratic Witt groups, or Lemma 3.1; those remain paper-level mathematics grounded in the cited exact sequence and finite-field theory. The Rust local decomposition computes the triple (ϕ_0, ψ, ϕ_1) with canonical uniformizers. It does not yet expose the global coordinate theorem as a public Witt-class API or implement ramified coordinate transport.

10 Conclusion

The nonsingular realization problem has a sharp two-layer answer. At the classification layer, the infinite unramified coordinate and all second residues give complete canonical coordinates for $W_q(\mathbb{F}_2(t))$. The wild odd-pole terms cannot be discarded, but finite trace characters present them basis-freely, and the games 0 and $*$ realize the resulting finite-support bit vector exactly. Reciprocity is one parity equation on the ramified Arf bits.

At the local-access layer, ramified Artin–Schreier normalization is the remaining theorem. It must be internalized by a refinement-independent arena rather than hidden in preprocessing. Beyond the nonsingular group, the quasilinear absorption relation proves that the original additive singular extension cannot exist. The obstruction is algebraic and precedes any choice of game rules.

References

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